INSURANCE CLAIMS ANALYTICS

From Raw Claims Data to Business Insight

An end-to-end analytics solution for fraud detection, claim performance, and underwriting reporting — built on your auto insurance claims portfolio.

CLIENT OUTCOME REVIEW • JULY 2026

THE BUSINESS CHALLENGE

1,000 Policies. No Visibility.

Fraud went undetected

No systematic way to see which regions or customer segments were driving fraudulent claims.

Approval outcomes unclear

No visibility into whether claim approvals lined up with the real risk being carried.

No cost trend visibility

Leadership couldn't see how claim costs were moving month to month, or where they were concentrated.

The data also carried quality gaps — **missing values, an unusual negative field, and no clear “was this claim approved or denied” outcome** — all handled and documented as part of this project.

WHAT WE BUILT

An End-to-End Analytics Solution

01

Structured Data Foundation

Raw spreadsheet data reshaped into a clean, connected database — customers, policies, vehicles, incidents, and claims.

02

Automated Data Pipeline

A repeatable Python process cleans, validates, and loads new data automatically, with built-in quality checks.

03

Deep Analytical Queries

24 analytical queries uncover fraud patterns, approval trends, high-value claims, and portfolio risk.

04

Executive Power BI Dashboard

A 4-page interactive dashboard gives leadership a live, self-service view of the entire claims book.

KEY OUTCOMES

Your Claims Book, At a Glance

1,000	24.7%	75.3%	$52,762
Total Claims Analyzed	Overall Fraud Rate	Claim Approval Rate	Average Claim Amount

These figures come directly from validated SQL analysis of the full 1,000-claim portfolio (Jan–Mar 2015).

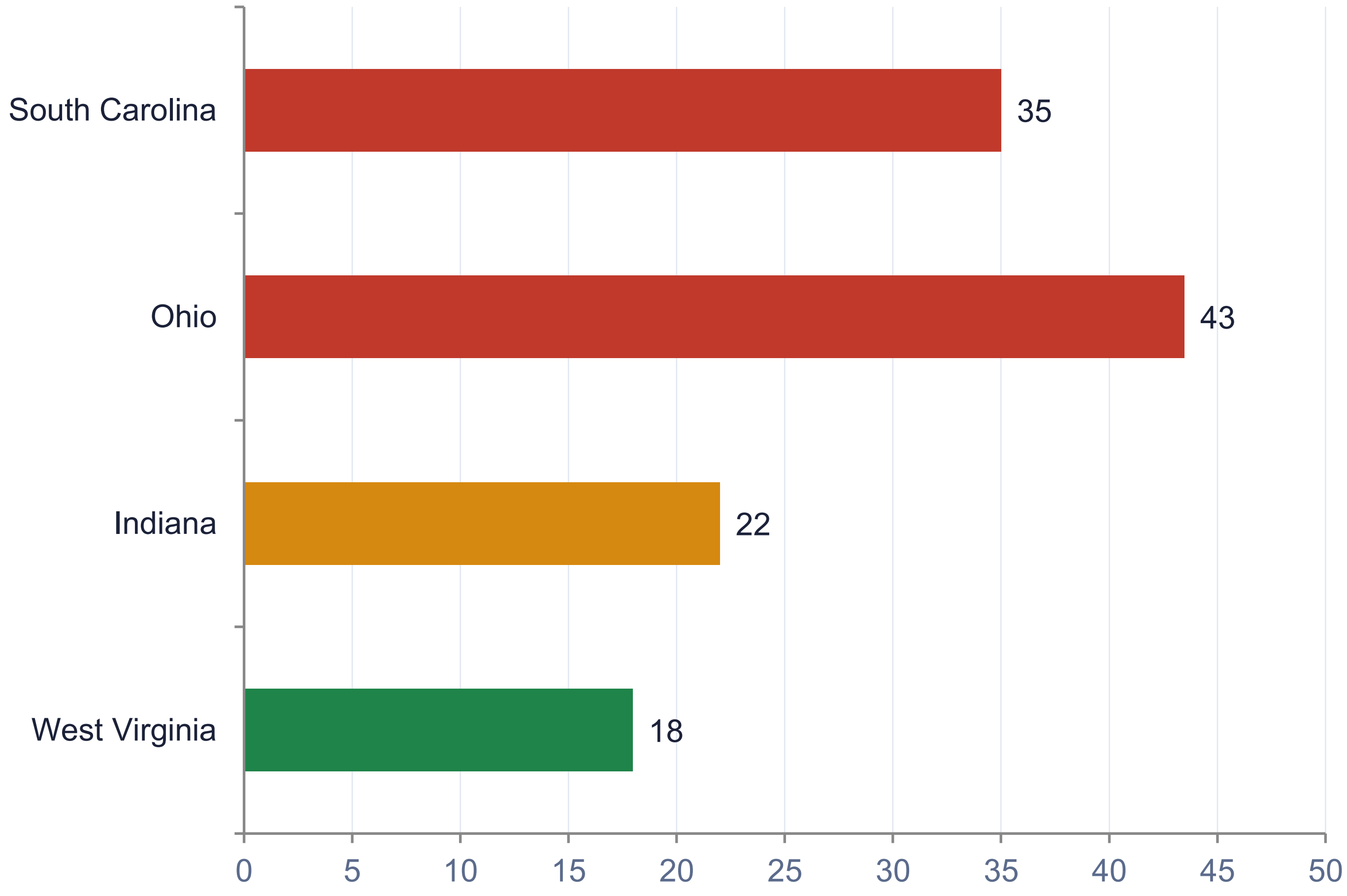
$41,813	$58,055	$70,593
25th Percentile	Median Claim	75th Percentile

Red / Amber / Green Status

Metric	Current Value	Status	What it means
Fraud Rate	24.7%	AMBER	Above the 20% comfort threshold — worth targeted review.
Approval Rate	75.3%	AMBER	Healthy, but below the 80% green benchmark.
Avg. Claim Amount	$52,762	AMBER	Mid-range against the $45K–$70K band.
Highest-Fraud Region	Ohio — 43.5%	RED	Well above portfolio average — priority investigation.
Lowest-Fraud Region	West Virginia — 18.0%	GREEN	Healthiest region in the book — a useful benchmark.

Thresholds were set from the actual spread in your data — not arbitrary limits — so each color reflects a real, defensible cutoff.

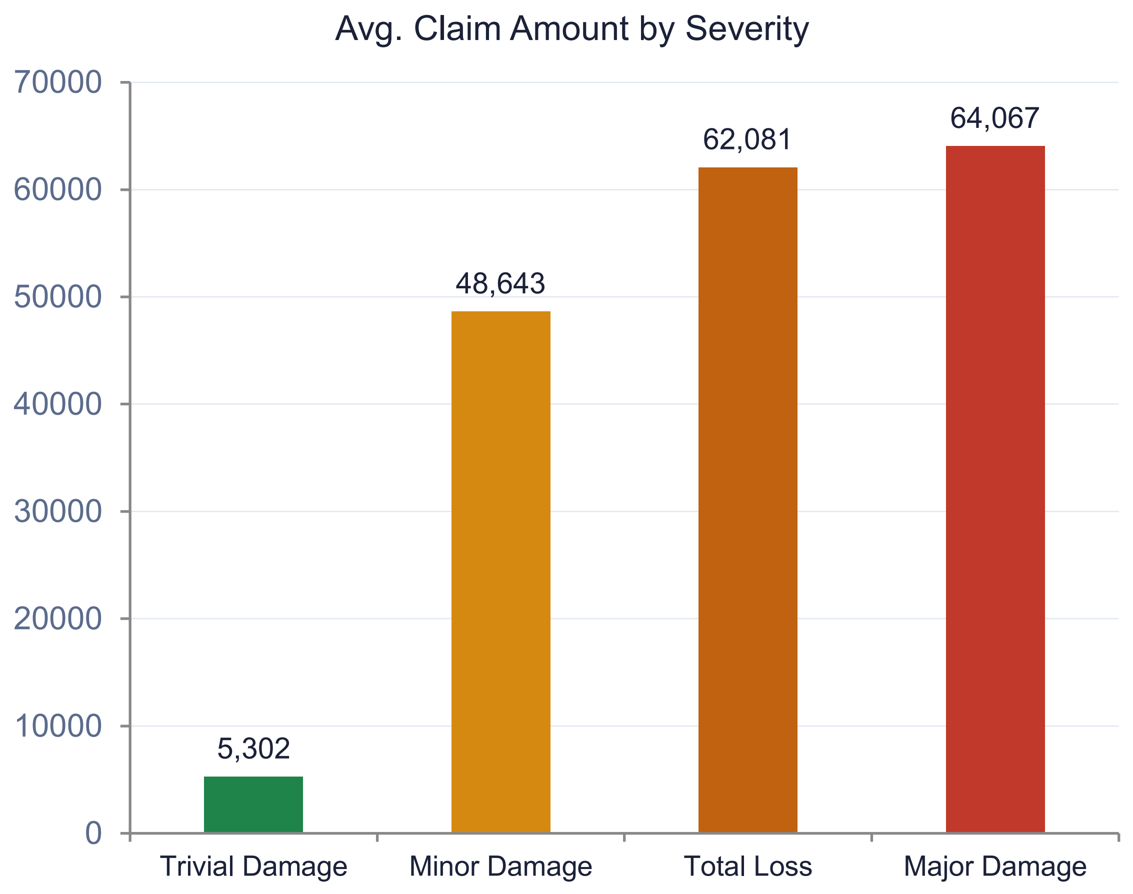
Fraud Risk Varies Sharply by Region



Why this matters

- We measured fraud by the state where the incident happened, not where the policy was written — this is where the real spread shows up (18% to 43%).
- Ohio and South Carolina stand out as elevated-risk regions and are good candidates for tighter claim review.
- West Virginia performs well and can serve as an internal benchmark for what “healthy” looks like.

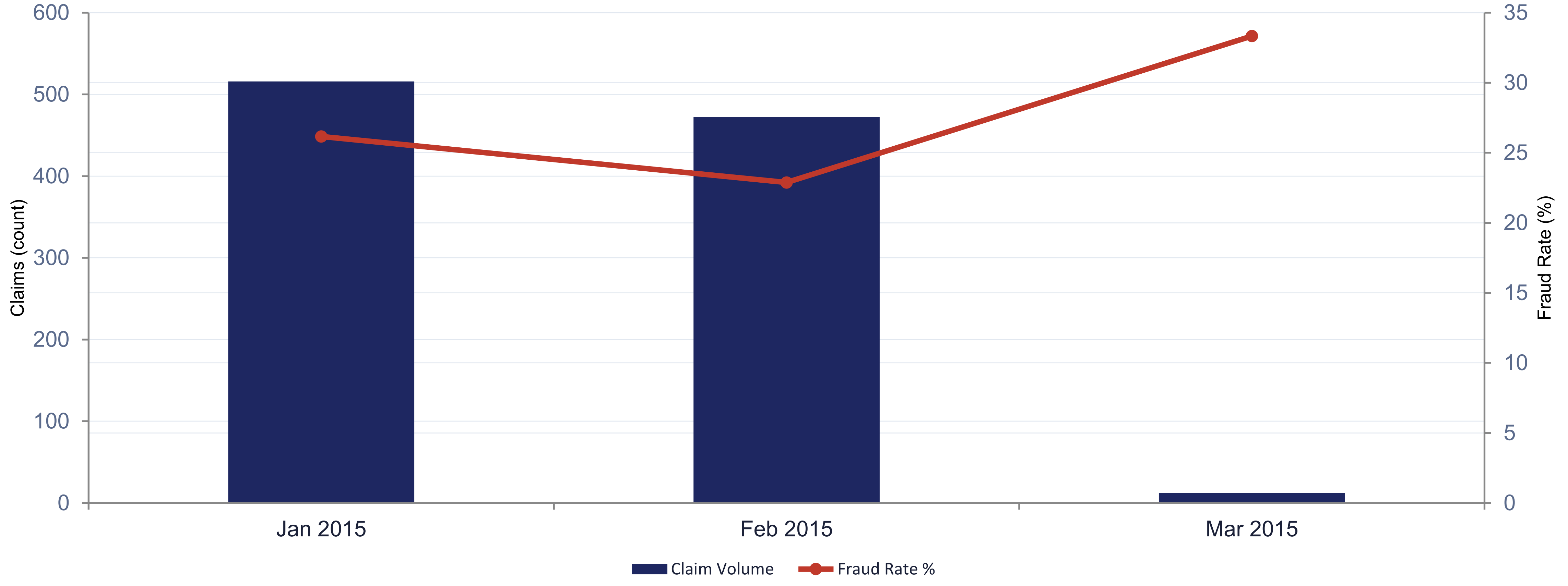
Severity Drives Both Cost and Approval



Severity	Approval Rate
Trivial Damage	93.3% – GREEN
Minor Damage	~78% – AMBER
Total Loss	~70% – AMBER
Major Damage	39.5% – RED

Major Damage claims cost the most on average and get approved the least — a natural focal point for adjuster review and fraud checks.

Monthly Claim Volume & Fraud Trend



Note: the source data covers only Jan–Mar 2015 (March is a partial month), so this trend is directional, not seasonal.

THE DELIVERABLE

A Live, Self-Service Power BI Dashboard

1. Executive Overview

KPI cards, RAG status, monthly trend, and a regional fraud map — the one-page view for leadership.

2. Fraud Analysis

State and occupation fraud-rate breakdowns, plus a top-20 high-value fraud claims table.

3. Claim Details (Drill-Through)

Click any region or claim to see the full customer-to-claim record, with one-click back navigation.

4. Portfolio / Underwriting View

Coverage-tier comparisons, age-band claim distribution, and claim-size mix for underwriting review.

OUR RECOMMENDATIONS

Turning Insight Into Action

Prioritize Ohio & South Carolina for fraud review

These regions run well above the portfolio's average fraud rate — start manual review and SIU (special investigation) triage here first.

Capture a true adjudication outcome going forward

Today's “approved/denied” status is estimated from fraud flags. Feeding in the real claims-system outcome will sharpen every metric.

Add extra scrutiny to Major Damage claims

This category has the highest average payout and the lowest approval rate — the highest-value place to tighten controls.

Extend the data window

Current data covers 3 months only. A full 12+ months will reveal real seasonality instead of short-term noise.

Use West Virginia as a benchmark

Its fraud and approval profile is the healthiest in the book — study what's different operationally and apply it elsewhere.

Roll the dashboard out to underwriting & claims teams

Self-service access means faster answers without waiting on ad-hoc reports.

Thank You

This project turned a flat spreadsheet into a structured, queryable data asset — and a live dashboard your teams can use every day.

Next Step: We're ready to walk your team through the live dashboard and discuss extending this to your full claims history.